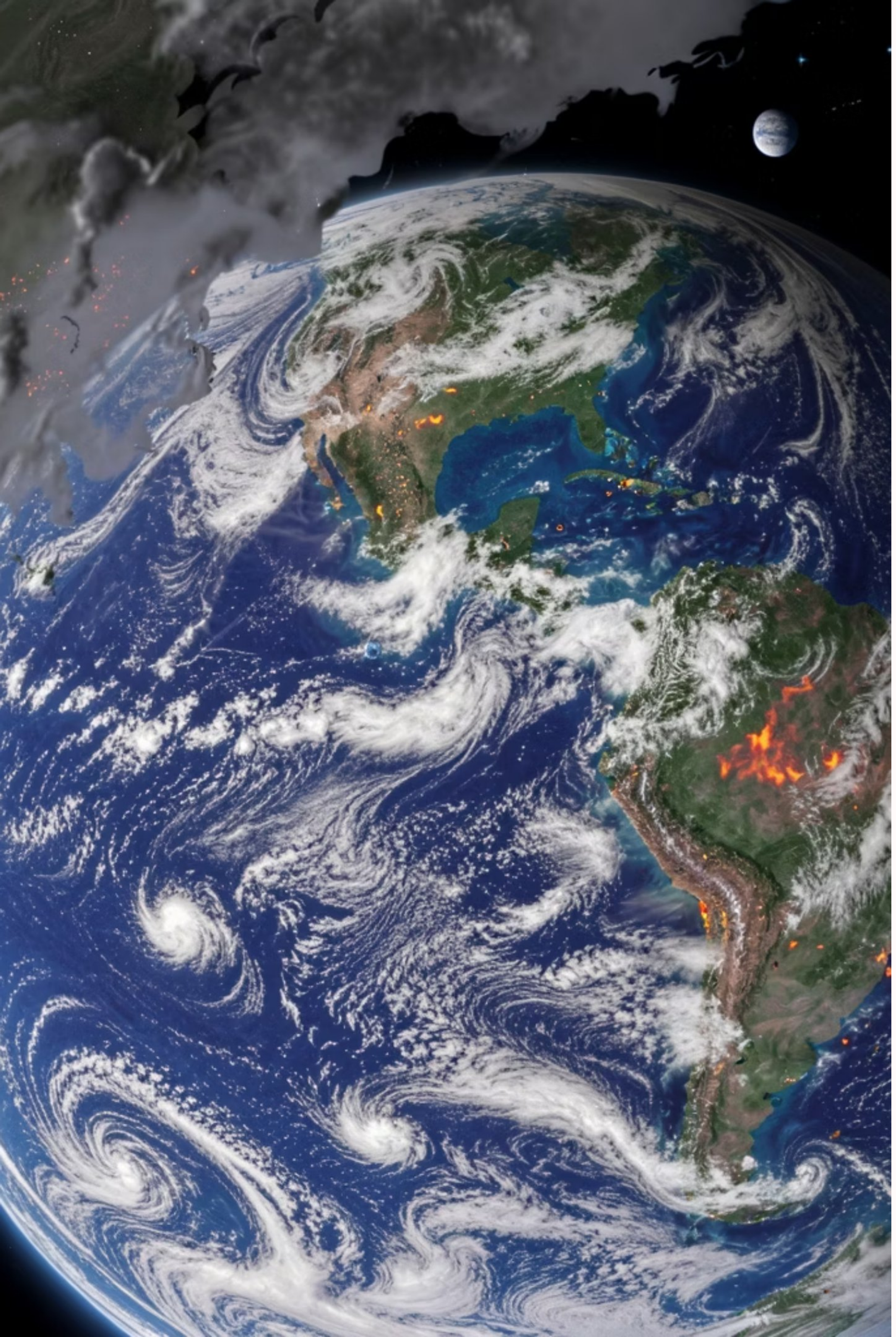


Hack It Out

Team: HACKnFORGE

PyClimaExplorer

Climate Data Dashboard: Visualizing Earth's Patterns

An interactive web dashboard for exploring climate datasets through intuitive visualizations

Project Objective

Interactive Exploration

Build a functional web dashboard that allows users to explore climate datasets stored in NetCDF format with ease

Visual Understanding

Enable users to visualize climate variables through graphical representations like heatmaps and time-series plots

Quick Insights

Help users quickly understand climate patterns and trends using intuitive, accessible visualizations

NetCDF Data Format

What is NetCDF?

Network Common Data Form is a widely used scientific data format for storing multidimensional climate and atmospheric measurements.

Key Characteristics:

- Stores multidimensional arrays efficiently
- Self-describing with metadata included
- Platform-independent and portable
- Supports large datasets seamlessly

Climate Variables in the Dataset

Temperature

Air and surface temperature measurements across different altitudes and time periods

Precipitation

Rainfall, snowfall, and other forms of water falling from the atmosphere

Wind Speed

Velocity and direction of atmospheric air movement at various pressure levels

Humidity

Water vapor content in the atmosphere affecting weather patterns

Pressure

Atmospheric pressure variations driving weather systems and circulation

Target Audience

Researchers

Climate scientists and researchers who need quick visualization tools to examine climate datasets without writing complex analysis code. This dashboard allows them to quickly inspect spatial distributions and temporal trends.

General Public

Educated citizens interested in understanding climate trends such as temperature changes or rainfall patterns in an easy and visual way, making climate data accessible to everyone.

Core Dashboard Functionality

01

Dataset Loading

Users can upload or load a NetCDF dataset into the dashboard. The system reads available variables such as temperature or precipitation.

02

Variable Selection

The user can choose which climate variable they want to visualize from the dataset with an intuitive interface.

03

Global Heatmap Visualization

A global heatmap displays the spatial distribution of a selected climate variable across the Earth's surface for a particular time.

04

Time-Series Line Plot

The time-series line plot shows how a specific climate variable changes over time at a selected geographic location.

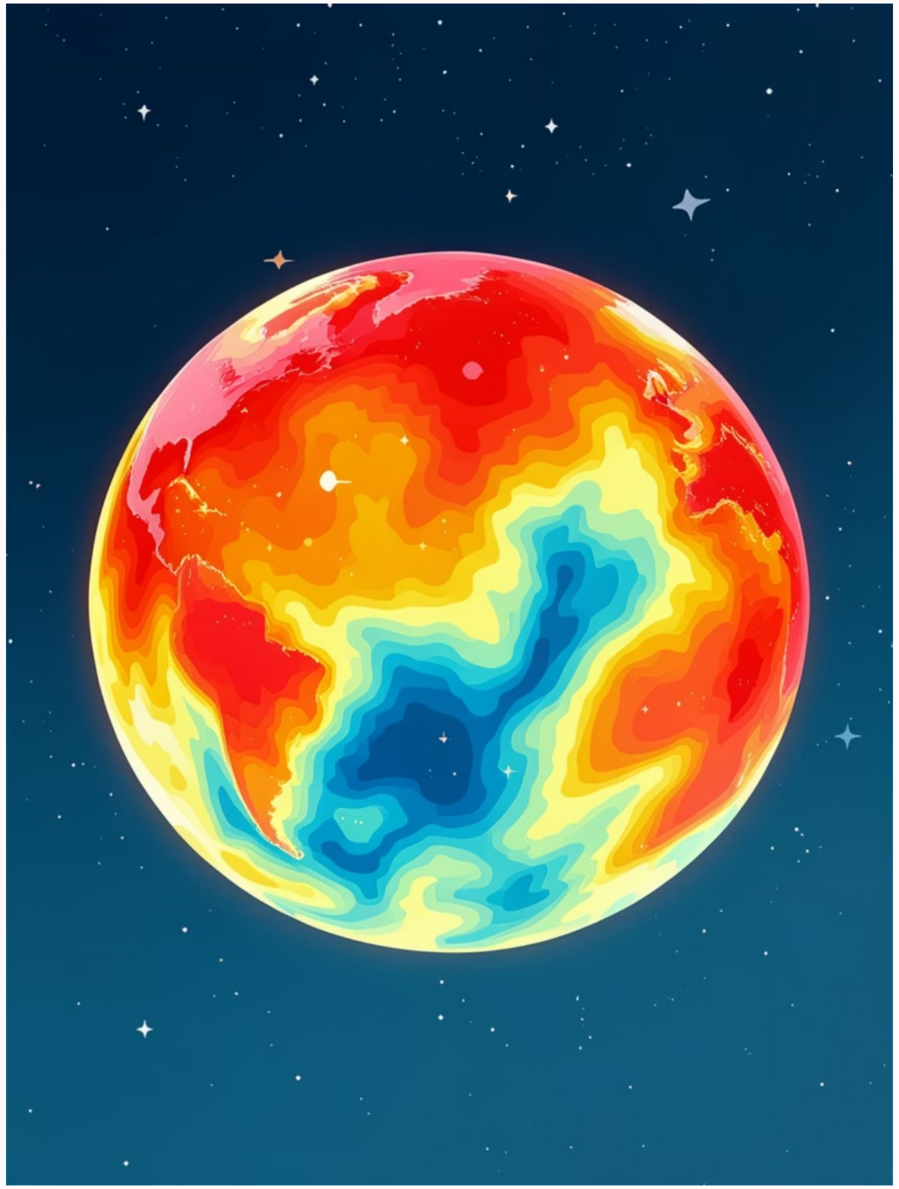
Visualizing Climate Patterns

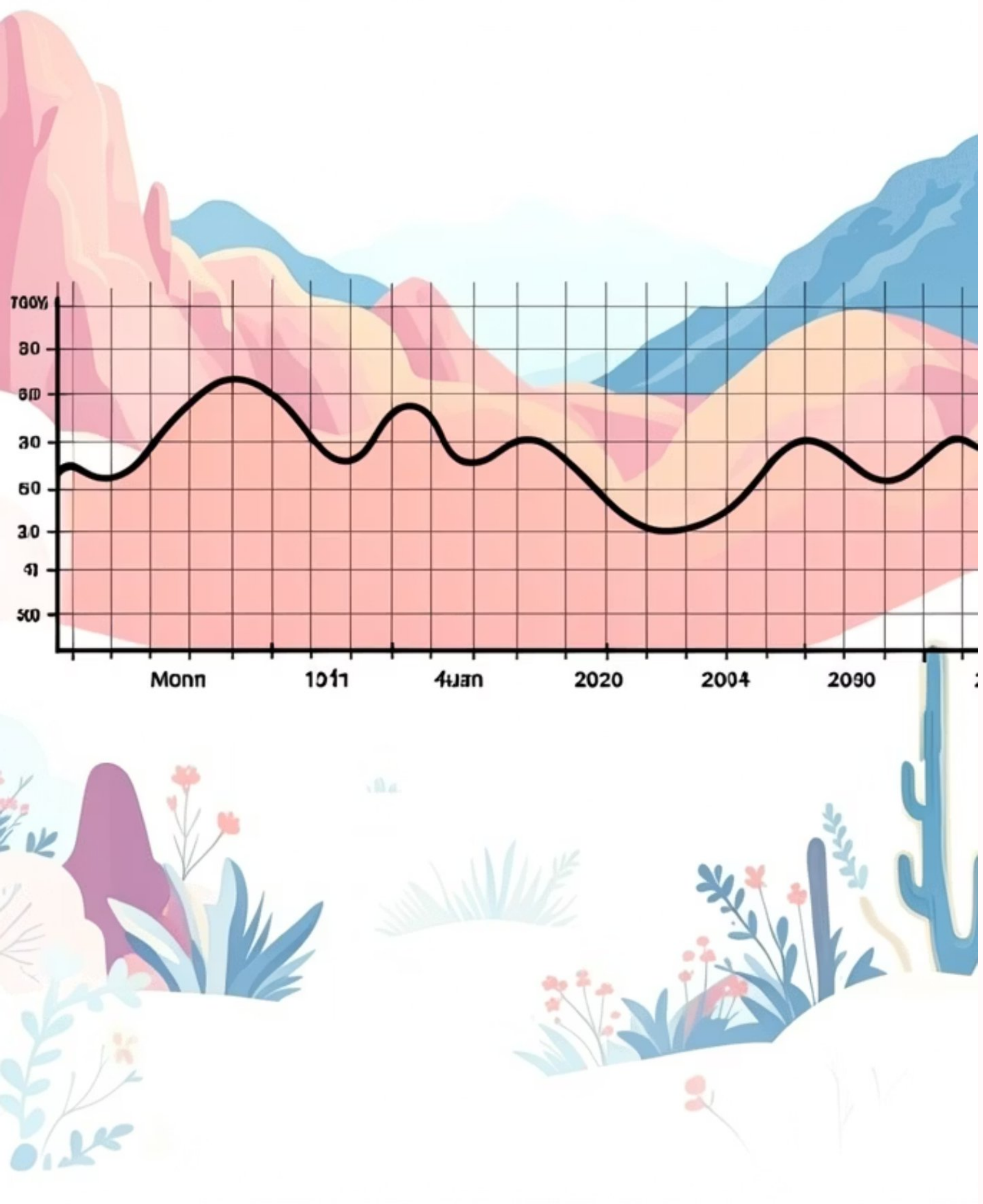
Global Heatmap

A global heatmap displays the spatial distribution of a selected climate variable across the Earth's surface for a particular time. The map uses colors to represent the magnitude of the variable.

- **Blue** represents lower values
- **Yellow/Green** represents moderate values
- **Red** represents higher values

This visualization helps users quickly identify geographic patterns such as hotter regions or areas with heavy rainfall.





Time-Series Line Plot Analysis

Location Selection

Users can choose a particular latitude and longitude on the globe to analyse.

Data Plotting

The system plots the values of the selected variable over the entire time dimension.

Trend Observation

Users can observe trends such as increasing temperatures or seasonal rainfall variations.

Technical Implementation

Streamlit

Creates the interactive web interface

xarray

Reads and processes NetCDF datasets

Plotly

Generates the visualizations

The dashboard workflow: User uploads a NetCDF dataset → System reads the dataset and extracts available variables → User selects a variable to visualize → Dashboard generates visualizations → Users interact by selecting time steps or locations.

Expected Outcomes

Accessible Climate Exploration

An interactive web dashboard that makes complex climate data accessible without advanced programming knowledge

Multi-Scale Visualization

Simultaneous spatial (global heatmap) and temporal (time-series plot) analysis capabilities

Flexible Data Support

Capability to load various NetCDF climate datasets from models like CESM or ERA5